Nonnegative and positive factorizations

Additivity for Cartesian-product slack matrices

Difficulty: extreme **Importance:** interesting to the community

Status: open; checked 2026-09-08
Area: nonnegative rank and polyhedral optimization

Context and notation

All factorizations are over the real numbers. For $X \in \mathbb{R}_{\geq 0}^{m\times n}$, define

$$\operatorname{rank}_+(X) = \min\{r \geq 0 : X = WH, \quad W \in \mathbb{R}_{\geq 0}^{m\times r},\ H \in \mathbb{R}_{\geq 0}^{r\times n}\}.$$

Problem statement

Let $P \subset \mathbb{R}^d$ and $Q \subset \mathbb{R}^e$ be full-dimensional polytopes with $d, e \geq 1$. Let $S \in \mathbb{R}_{\geq 0}^{m\times n}$ and $T \in \mathbb{R}_{\geq 0}^{p\times q}$ be their slack matrices: rows correspond to all facets in irredundant inequality descriptions, columns to all vertices, and each entry is the right-hand side minus the left-hand side of that facet inequality at that vertex. Construct $C \in \mathbb{R}_{\geq 0}^{(m+p)\times nq}$ by giving it one column for each pair (i, j) of vertices:

$$C[:, (i, j)] = \begin{pmatrix} S[:, i] \\ T[:, j] \end{pmatrix}.$$

Question

Is

$$\operatorname{rank}_+(C) = \operatorname{rank}_+(S) + \operatorname{rank}_+(T)$$

always true? The hypotheses that S, T are polytope slack matrices are part of the question. Through the slack-factorization theorem, this also asks whether the minimum number of inequalities in an extended formulation is additive under Cartesian products.

References

Gillis, *Nonnegative Matrix Factorization*, §3.6.5, pp. 93–94. Hans Raj Tiwary, Stefan Weltge, and Rico Zenklusen, *Extension complexities of Cartesian products involving a pyramid*, Information Processing Letters **128** (2017), 11–13, introduction and main theorem.

Status evidence

The equality is proved when at least one factor is a pyramid. Searches for Cartesian-product extension-complexity additivity, counterexamples, and 2025–2026 developments found no general resolution. Stefan Weltge poses the general equality again in the Cargèse workshop open problems, September 20, 2022, p. 1. This is the latest explicit general open-status confirmation located; its age is a limitation of this entry.